HV VERTEX ELECTRONICS

Frontend Website Project Charter

A complete build brief for AI / vibe-coding platforms (Lovable, Bolt, v0, Cursor, Replit, etc.) to generate a professional, production-ready website for HV Vertex Electronics — with zero missing context.

"Build. Connect. Automate. Simplify."

Prepared for: HV Vertex Electronics | Document type: Website Project Charter v1.0

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1 How To Use This Document

Read this first — it explains how to hand this charter to an AI coding tool.

This charter is written to be pasted directly into any AI/vibe-coding platform (Lovable, Bolt.new, v0, Cursor, Replit Agent, Claude, etc.) as the single source of truth for building the HV Vertex Electronics frontend website. It contains the company's real content, product list, sitemap, section-by-section page specs, design system, and functional requirements — so the AI does not need to guess or invent facts.

- Upload/paste this entire PDF (or its text) as the first prompt, then say: "Build the complete website exactly as specified in this project charter."
- Upload the company logo and any product photos separately and reference them (e.g. "use the uploaded logo in the navbar and footer").
- If the tool allows multiple prompts, build page-by-page in the order given in Section 5 (Sitemap), using the section breakdowns in Section 6.
- Keep the Design System (Section 7) consistent across every page and prompt.

2 Project Overview

Project name	HV Vertex Electronics — Corporate & Product Website
Project type	Frontend marketing / brochure website (single company site, expandable to e-commerce later)
Primary goal	Present HV Vertex as a credible, modern electronics & automation company; showcase product categories; generate enquiries via contact/WhatsApp; support students, engineers & industries
Primary audience	Makers, students & engineering colleges, hobbyist/IoT developers, small industries & workshops, home-automation customers, B2B bulk buyers
Tone	Confident, practical, tech-forward, approachable — not overly corporate, not childish
Platform target	Responsive website (desktop, tablet, mobile) — built via AI/vibe-coding tool
Phase 1 scope	Static/frontend informational site with working contact form + WhatsApp click-to-chat (no login, no payment gateway yet)

3 Company Background

Use this content verbatim/near-verbatim across the site — do not invent new facts.

3.1 About the Company

HV Vertex Electronics is an emerging electronics and technology company focused on developing, manufacturing, assembling, and supplying electronic components, development boards, smart systems, automation solutions, electrical products, and hardware accessories. The company works across Arduino development boards, ESP32-based systems, embedded electronics, IoT devices, home automation, smart appliances, electrical components, sensors, modules, connectors, screws, fasteners, and other electronic and hardware products. The goal is not limited to selling individual components — HV Vertex designs complete technology solutions that solve real-world problems.

3.2 Our Approach — Problem to Solution

HV Vertex follows a problem-to-solution approach instead of developing technology for its own sake:

Problem	HV Vertex Solution
A fan needs to be manually switched ON/OFF	Automatic temperature-based fan controller
Lights are frequently left ON unnecessarily	Sensor-based or scheduled automatic lighting
Users want to control appliances remotely	Wi-Fi / Bluetooth-enabled smart controllers
Traditional appliances aren't connected to smart systems	Retrofit automation modules for existing appliances

3.3 Vision

"To build a smarter, easier, and more connected future through accessible electronics and intelligent automation." HV Vertex envisions becoming a trusted electronics and smart-technology company, bringing together electronics, IoT, automation, embedded systems, electrical technology, and intelligent machines for homes, businesses, educational institutions, and industries.

3.4 Mission

To design, develop, manufacture, and supply reliable and affordable electronics and smart systems that solve real-world problems, including:

- Develop quality Arduino and ESP32-based products
- Build practical IoT and automation systems
- Make smart technology affordable and accessible
- Develop innovative household automation products
- Reduce unnecessary energy consumption through automation
- Support students, makers, engineers, and developers
- Provide reliable electronic and electrical components
- Combine electronics, software, mechanical systems, and automation into complete solutions

3.5 Core Philosophy & Promise

"Technology should make life easier, not more complicated." HV Vertex is committed to combining electronics, automation, IoT, embedded systems, electrical products, mechanical hardware, and smart-system development under one platform — helping customers build, automate, connect, and simplify. HV Vertex also promises best-quality, affordable-price products, fast delivery, and warranty assurance on all products.

4 Company Goals & Future Scope

Use as content for the About / Vision / Future pages.

4.1 Our Goals

- Product Development — a strong portfolio of Arduino, ESP32, sensor, controller, power, electrical, and automation products.
- Smart Home Innovation — affordable automatic fans, smart lighting, appliance controllers, security systems, energy management.
- Everyday Automation — automate repetitive household/workplace tasks to save time and effort.
- Energy Efficiency — intelligent systems that reduce energy usage via schedules and sensor conditions.
- Quality & Reliability — high standards in design, components, manufacturing, testing, and support.
- Innovation — ongoing R&D; in IoT, embedded systems, AI, robotics, sensors, wireless comms, power electronics.
- Customer-Centered Products — designed around real customer requirements.
- Build a Complete Ecosystem — one trusted source for components, boards, controllers, smart appliances, and complete solutions.

4.2 Future Scope

HV Vertex plans to expand beyond boards and components into complete smart products, robotics, and industrial automation:

Area	Details
Robotics Development	Educational, autonomous, sensor-based, industrial, IoT-enabled and remote-controlled robotic systems; custom robotics projects.
Industrial Automation	Industrial controllers, automated machine systems, motor-control, sensor-based monitoring, IoT-based industrial monitoring, process-control systems.
Support for Students & Institutions	Boards, components, sensors, robotics parts, project hardware, embedded-system solutions, prototype development, and academic project assistance.
Research & Development	Continuous R&D; in embedded systems, IoT, robotics, AI, industrial automation, smart home tech, and energy management.
Innovation & Custom Solutions	Custom electronics/automation combining Electronics + Embedded Systems + Software + IoT + Robotics + Automation + Mechanical Systems.

5 Product Categories (Full Content for Products Page)

Use this exact list to build the Products / Categories page. Each item below should become a product-category card with icon, title, and bullet description.

1. Microcontrollers & Processors

- Different types of microcontrollers and processors
- Small and large processors in different sizes and specifications
- Bluetooth and non-Bluetooth options
- Our own branded microcontrollers and processors

2. Our Own Programming App

- Easy-to-use programming app for microcontrollers and processors
- Supports Arduino, ESP, and our own branded hardware
- Program and upload code directly to microcontrollers
- Includes an AI agent to assist with programming, algorithms, error identification, and code improvement

3. PLC

- PLC for small-scale industrial applications
- Suitable for industrial automation and control projects

4. Motor Drivers

- DC motor drivers
- BLDC (Brushless DC) motor drivers
- 20A and 60A motor driver options
- Suitable for robotics and other motor-control applications

5. Home Automation Products

- Light control, fan control, AC control
- Home appliance control
- Smart and automated home-control products

6. Robotics Products & Components

- Electronic and mechanical components
- Robotics-related products
- Components for different robotics applications

7. Screws, Bolts & Fasteners

- Different types and sizes of screws, bolts and nuts
- Fasteners for robotics and mechanical applications
- Suitable for industrial and general-purpose applications

8. Customized Products

- Customized microcontrollers and processors
- Customized motor drivers
- Robotics and automation products
- Hardware and software based on customer requirements

Note: state clearly on the site — "These are our current product categories and products" — since the catalog is expected to grow.

6 Sitemap — Pages To Build

#	Page	Purpose / Key Content
1	Home	Hero, quick intro, category highlights, why-choose-us, featured problem→solution examples, CTA
2	About Us	Company background, approach, vision, mission, philosophy, promise
3	Products	All 8 product categories with descriptions (from Section 5)
4	Solutions / Automation	Problem→Solution examples, home automation & IoT use-cases
5	Robotics & Industrial Automation	Future scope details for robotics and industrial automation
6	For Students & Institutions	How HV Vertex supports colleges, engineering students and makers
7	Research & Development	R&D; focus areas and future vision
8	Contact Us	Contact form, phone numbers, email, WhatsApp button, map placeholder

Small-site variant: pages 4–7 can be combined into About/Vision sections if a single-page or 4-page site is preferred — flag this to the AI tool if a shorter build is wanted.

7 Page-by-Page Section Breakdown

Exact section order the AI tool should generate for each page.

7.1 Home Page

- **Navbar** — logo (left), links: Home / About / Products / Solutions / Students / Contact, phone number + WhatsApp icon (right), sticky on scroll
- **Hero section** — bold headline ("Build. Connect. Automate. Simplify."), sub-line from the vision statement, two CTA buttons ("Explore Products" / "Contact Us"), circuit-board / IoT themed background graphic
- **Trust strip** — 3–4 short badges: Best Quality & Affordable Price, Fast Delivery, Warranty Assurance, Own-Branded Products
- **About preview** — 2–3 line company summary + "Learn more" link to About page
- **Product categories grid** — 8 cards (icon + title + 1-line desc) from Section 5, each linking to Products page
- **Problem → Solution showcase** — 4 cards using the table in Section 3.2, presented as "Problem" vs "Our Solution"
- **Why Choose HV Vertex** — Quality & Reliability, Innovation, Customer-Centered, Ecosystem approach (from Section 4.1)
- **For students / makers callout** — short banner inviting students & engineers to explore project support
- **Future scope teaser** — 1 short section teasing Robotics / Industrial Automation / R&D; with "Read more" link
- **CTA banner** — "Have a project in mind? Let's build it together" + Contact button
- **Footer** — logo, tagline, quick links, product category links, contact details (9182503188, 91827 36781, hvvertex2026@gmail.com), social icons placeholders, copyright

7.2 About Us Page

- Header banner with page title
- Full "About the Company" text (Section 3.1)
- "Our Approach" section with the 4 problem→solution examples as visual cards
- Vision statement in a highlighted quote block
- Mission — bullet list (Section 3.4)
- Core Philosophy quote block
- Our Promise closing statement + tagline

7.3 Products Page

- Page header + short intro line: "We are developing and selling products in the following categories"
- Filter/tab bar (optional, static for phase 1): All / Boards & Processors / Automation / Robotics / Hardware
- 8 product-category cards in a responsive grid, each with icon, name, and full bullet list from Section 5
- Highlight callout on the Programming App + AI agent feature (differentiator)
- Note banner: "These are our current product categories — more coming soon"
- CTA: "Need a custom solution? Contact us" linking to Contact page

7.4 Solutions / Automation Page

- Intro explaining the problem-to-solution philosophy
- Detailed cards for: Home Automation (light/fan/AC/appliance control), Energy Efficiency, Remote Control via Wi-Fi/Bluetooth, Retrofit Automation Modules
- "How it works" simple 3-step visual: Identify problem → Design solution → Deploy smart system

7.5 Robotics & Industrial Automation Page

- Two-column layout: Robotics Development vs Industrial Automation (content from Section 4.2 table)

- Bullet lists for each: robotics systems supported / industrial automation focus areas
- Future Vision closing paragraph

7.6 For Students & Institutions Page

- Intro: HV Vertex as a technology partner for colleges, engineering students and project developers
- What we provide list: boards, components, sensors, robotics parts, project hardware, embedded solutions, prototyping support, learning resources, academic project assistance
- Closing statement about moving from theoretical learning to practical experimentation
- CTA: "Get project support" button linking to Contact

7.7 Research & Development Page

- R&D; focus areas list (Embedded systems, IoT, Robotics, AI, Industrial automation, Smart home tech, Sensors, Wireless comms, Power electronics, Energy management)
- Innovation & Custom Solutions section (the six-discipline integration statement)
- Future Vision paragraph as closing statement

7.8 Contact Page

- Contact form: Name, Email, Phone, Subject, Message, Submit button
- Direct contact block: phone numbers (9182503188 / 91827 36781), email (hvvertex2026@gmail.com)
- Click-to-chat WhatsApp button
- Map placeholder (embed later once address is available)
- Business hours placeholder (to be confirmed by HV Vertex)

8 Design System (Give This To The AI Tool As-Is)

8.1 Visual Direction

Modern, clean, tech/electronics-industry aesthetic — think circuit boards, IoT dashboards and automation control panels rather than a generic corporate template. Confident dark navy + electric blue base with cyan and amber accents suggesting energy, connectivity and precision engineering.

8.2 Color Palette

Role	Hex	Usage
Primary — Deep Navy	#0B1E3A	Header/footer backgrounds, dark sections
Primary — Electric Blue	#0F6FFF	Buttons, links, key accents, headings
Accent — Cyan	#00C2CB	Secondary accents, icons, hover states
Accent — Amber	#FF9E1B	Highlights, energy/automation callouts, CTAs (used sparingly)
Neutral — Dark Grey	#33404F	Body text
Neutral — Light Grey	#F1F4F8	Section backgrounds, card backgrounds
Base — White	#FFFFFF	Primary background, card surfaces

8.3 Typography

Headings	Sans-serif, geometric/technical feel — e.g. Poppins, Space Grotesk, or Inter (Bold/SemiBold)
Body text	Inter, Roboto, or system-ui sans-serif — Regular/Medium, 16px base on desktop
Hierarchy	H1 32–44px, H2 26–32px, H3 20–24px, Body 16px, Small/labels 13–14px

8.4 UI Style Notes

- Rounded-corner cards (8–12px radius) with soft shadows for product/category cards
- Subtle circuit-line / dot-grid or PCB-trace pattern as background texture in hero and footer
- Icon set: line-style tech icons (microchip, Wi-Fi, sensor, gear/motor, robot arm, plug, lightbulb)
- Buttons: solid electric-blue primary, outline/ghost secondary, amber for high-priority CTAs only
- Generous white space; avoid clutter despite the large product list
- Micro-animations on scroll (fade/slide-up) and hover states on cards/buttons — keep subtle, not flashy
- Fully responsive: desktop, tablet, and mobile breakpoints; mobile nav collapses to a hamburger menu

8.5 Logo & Imagery

- Company logo will be supplied separately by HV Vertex — place in navbar (left) and footer, and use as favicon
- Until real product photography is supplied, use clean placeholder imagery/illustrations of circuit boards, microcontrollers, IoT devices and smart-home scenes that match the palette above
- Do not use any third-party branded product images without permission

9 Functional & Technical Requirements

9.1 Functional Requirements

- Responsive layout — works cleanly on mobile, tablet and desktop
- Sticky/fixed navigation bar with smooth-scroll or routed navigation between pages
- Working contact form (name, email, phone, subject, message) — connect to email or a form backend (e.g. Formspree, EmailJS, or a serverless function) since this is a frontend-only phase
- Click-to-chat WhatsApp button (wa.me link) using the provided mobile numbers
- Clickable phone numbers (tel:) and email (mailto:) links
- Product category cards with hover states; optional simple filter/search bar for later phases
- SEO basics: page titles, meta descriptions, alt text on all images, semantic HTML headings
- Fast load performance — optimized/compressed images, lazy loading below the fold
- Accessible color contrast and keyboard-navigable menus
- 404 page and a simple loading state for form submissions

9.2 Suggested Tech Stack

Framework	React (Next.js) or plain HTML/CSS/JS — either works well with vibe-coding platforms
Styling	Tailwind CSS (fast to theme with the palette in Section 8) or styled-components
Icons	Lucide, Phosphor, or Heroicons (line-style tech icons)
Forms	Formspree / EmailJS / simple serverless function for contact form submissions
Hosting (phase 1)	Vercel, Netlify, or GitHub Pages for a static/frontend deployment

9.3 Content Rules For The AI Tool

- Do not invent new company facts, prices, addresses, or team member names — none were provided
- Keep all figures (20A/60A motor drivers, contact numbers, email) exactly as given in this charter
- Leave clearly marked placeholders for: business address, business hours, social media links, and product photography until HV Vertex supplies them
- Use "HV Vertex Electronics" as the full brand name on first mention per page, "HV Vertex" thereafter

10 Ready-to-Paste Master Prompt

Copy everything in the box below directly into your AI/vibe-coding platform after uploading this PDF and the logo file:

> *Build a complete, responsive, multi-page frontend website for "HV Vertex Electronics" using the attached Project Charter PDF as the single source of truth for all company facts, product categories, vision/mission text, and page content. Follow the Sitemap (Section 6) and Page-by-Page Section Breakdown (Section 7) exactly for structure. Apply the Design System in Section 8: deep navy (#0B1E3A) and electric blue (#0F6FFF) as primary colors, cyan (#00C2CB) and amber (#FF9E1B) as accents, clean sans-serif typography (Inter/Poppins), rounded cards, subtle circuit/PCB-pattern textures, and smooth scroll animations. Implement the Functional Requirements in Section 9: responsive nav, working contact form, WhatsApp click-to-chat, clickable phone/email links, SEO meta tags, and optimized images. Use the uploaded logo in the navbar and footer. Do not invent facts not present in the charter — leave placeholders for address, hours, and social links.*

11 Recommendations For Further Development

Ideas beyond Phase 1 — for the roadmap.

Phase 2 — E-Commerce Layer

- Add a full online store: cart, checkout, payment gateway (Razorpay/Stripe), order tracking
- Product detail pages with specs, datasheets (PDF downloads), stock status, and pricing
- Bulk/B2B enquiry pricing tier for industries and institutions

Phase 2 — AI Agent Integration

- Surface the company's own "AI agent" (from the Programming App) as a public chatbot widget for code help, product recommendations, and troubleshooting
- This is a strong differentiator — highlight it prominently once available on the live site

Phase 2 — Learning & Community Hub

- Blog / tutorials section for Arduino, ESP32, and IoT project guides — great for SEO and student engagement
- Downloadable project guides, wiring diagrams, and starter code
- Community forum or Discord/WhatsApp community link for makers and students

Phase 2 — Customer Trust & Proof

- Testimonials / case studies once available
- Certifications, quality standards, or lab-testing badges
- Real product photography and short demo videos of automation products in action

Phase 3 — Accounts & Dashboards

- Customer login for order history, saved carts, and support tickets
- B2B/institution portal for colleges ordering project kits in bulk
- Admin dashboard for HV Vertex to manage products, orders, and enquiries

Phase 3 — Robotics & Industrial Microsites

- Dedicated sub-sites or landing pages for Robotics and Industrial Automation as those product lines mature
- Case studies for industrial clients showing automation ROI (energy saved, downtime reduced)

Ongoing — SEO & Marketing

- Target long-tail keywords: "ESP32 home automation India", "BLDC motor driver 60A", "engineering project components"
- Google Business Profile once a physical address/showroom is available
- Email newsletter signup for new product launches

Ongoing — Performance & Ops

- Analytics (Google Analytics / Plausible) to track which product categories get the most interest
- Live chat or WhatsApp Business API for faster response to enquiries
- Multi-language support (English/Tamil) given the local customer base

End of Project Charter — HV Vertex Electronics

Build. Connect. Automate. Simplify.